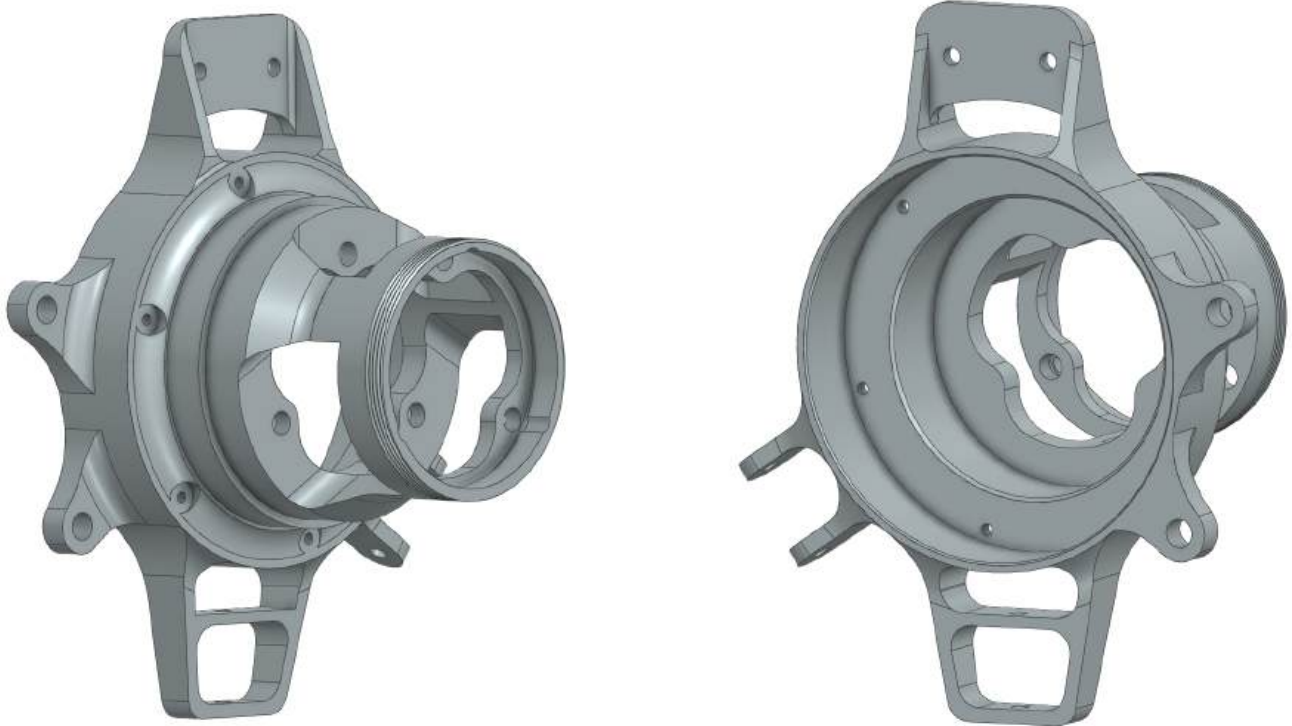
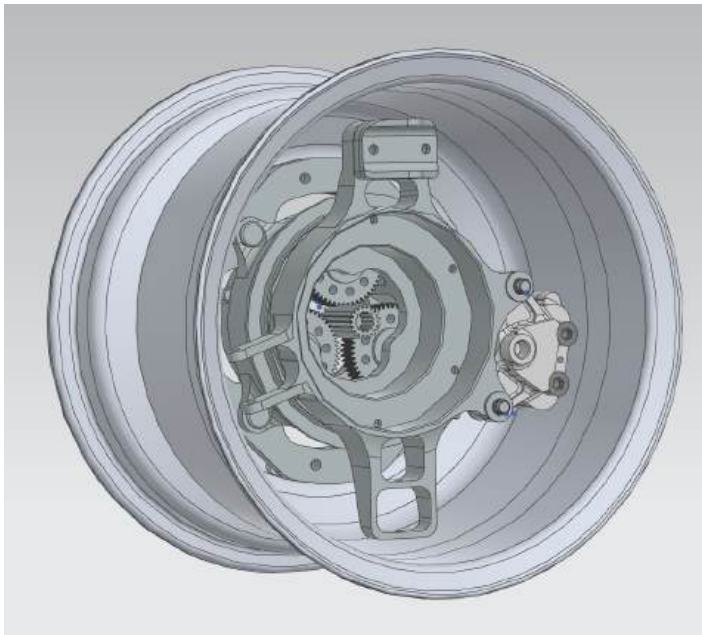


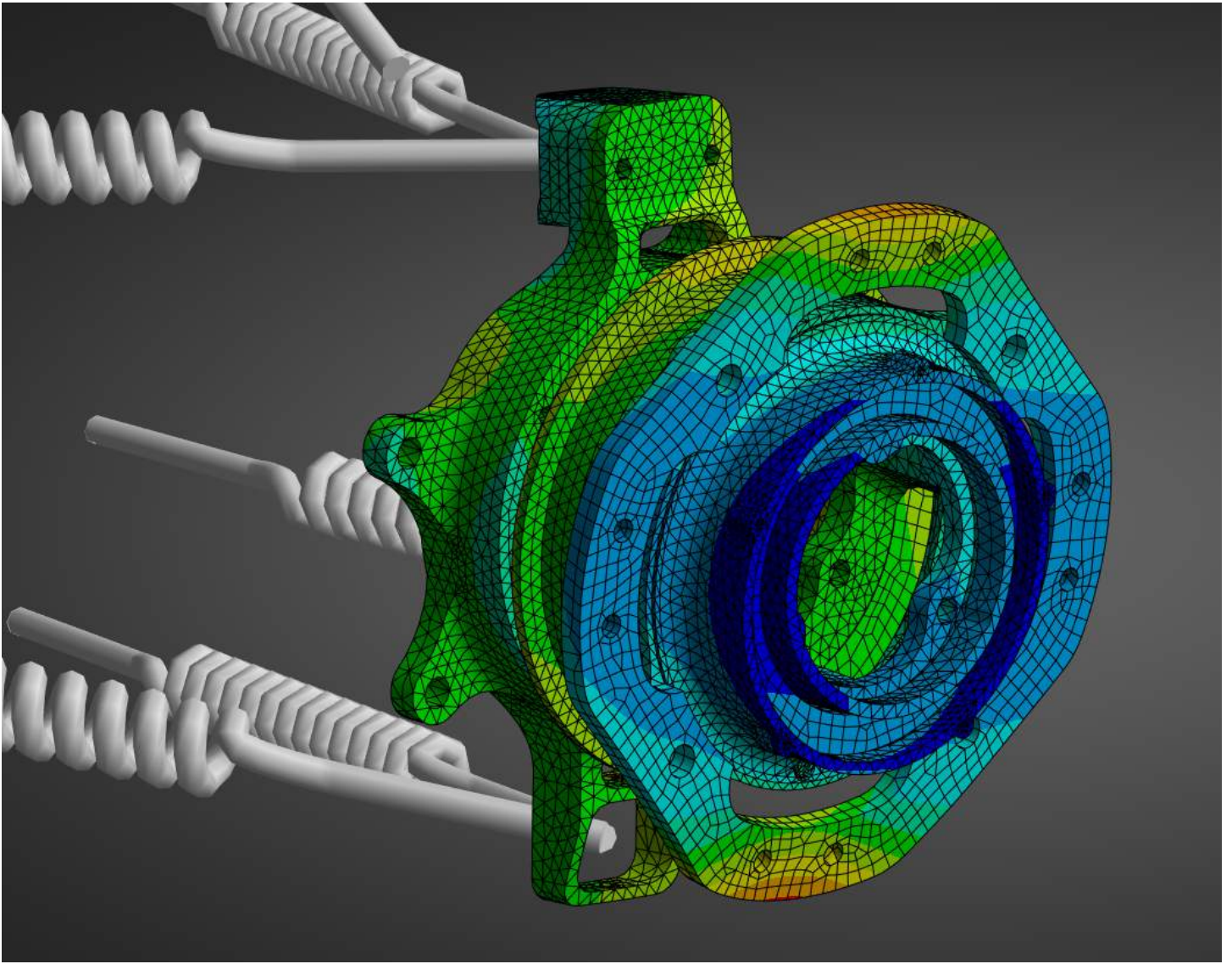
Purdue Electric Racing - Outrunner Upright Design



Unibody Upright & Planet Carrier Design

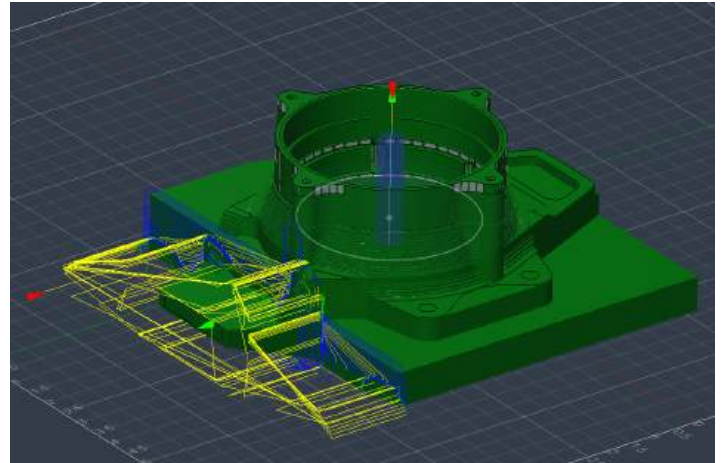
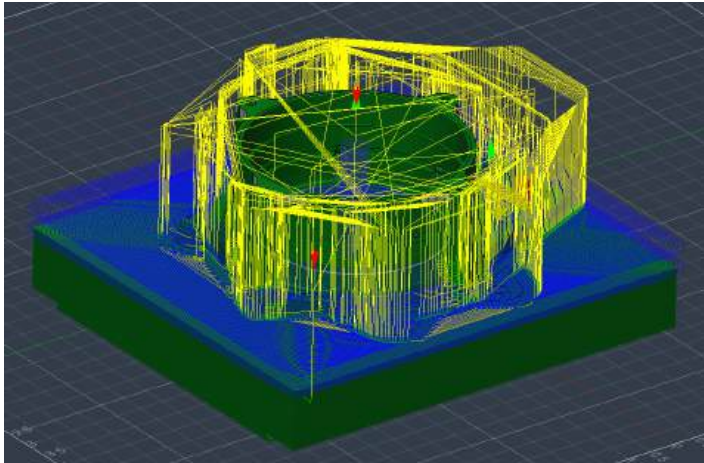


Full single wheel drivetrain assembly (without brake rotor, motor and accompanying peripheral parts)

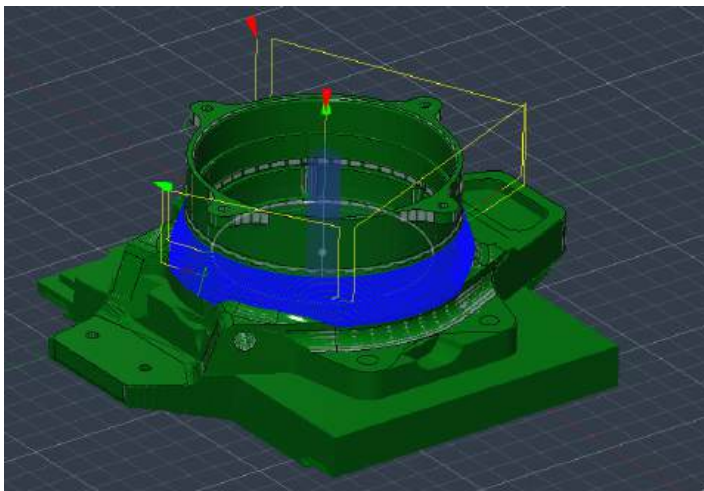


Rear Upright & Hub Group FEA

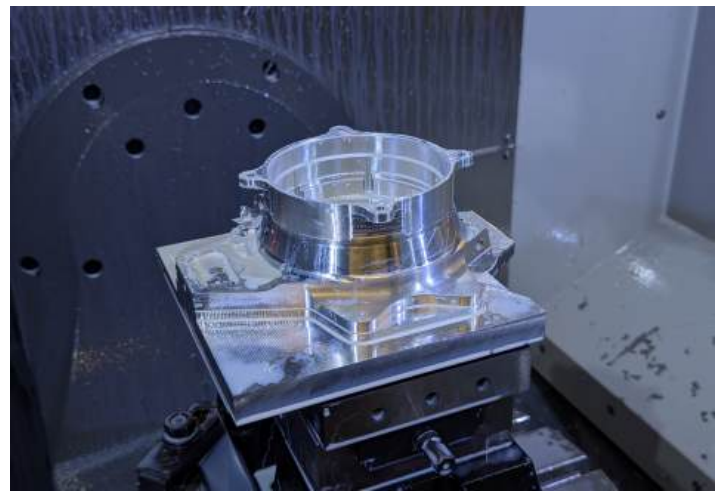
Purdue Electric Racing - Wheel Upright Manufacturing



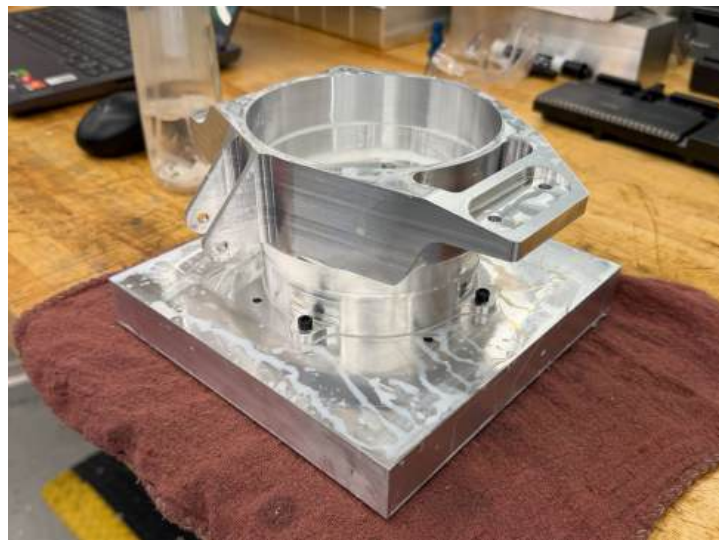
^ 5-Axis Roughing CAM Toolpaths for Rear Wheel Upright ^



^ 5-Axis Finishing Flow Operation ^



^ Upright After Completed 5-Axis Operation ^

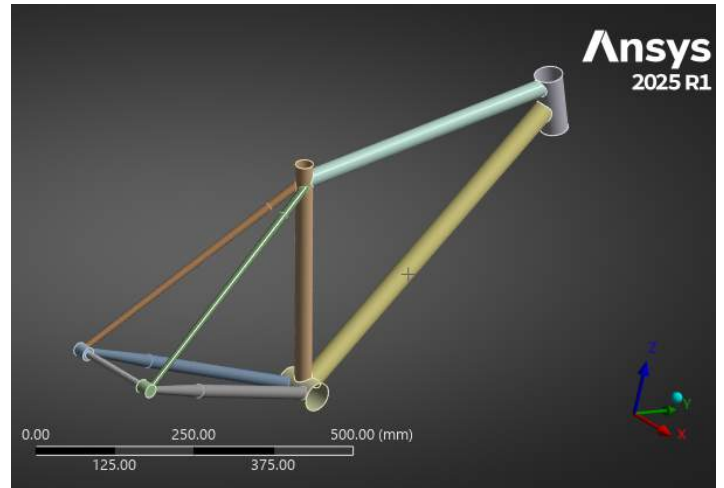


^ Upright After Final 3-Axis Operation ^

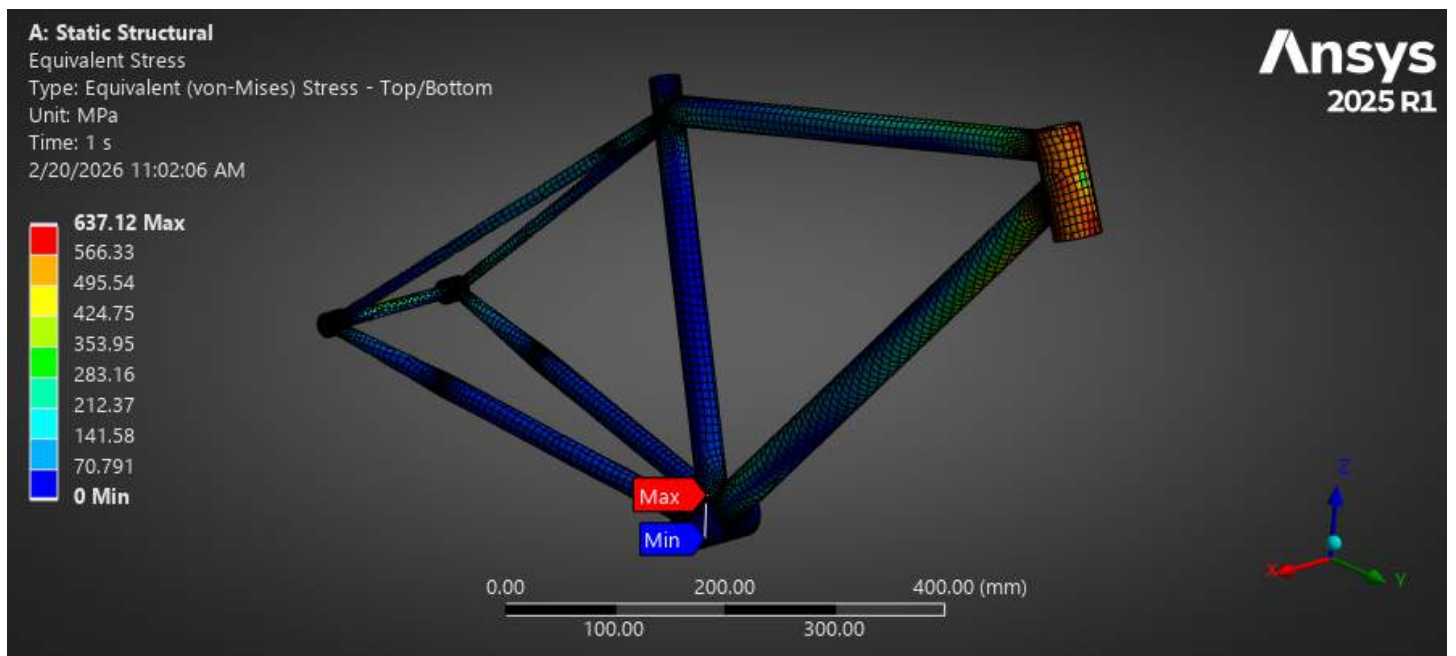
Personal Project: Custom Electric Bike



^ Top-Level Frame Assembly ^



^ Frame Prepared for Meshing in Ansys ^



^ Von-Mises Equivalent Stress Results in Ansys ^

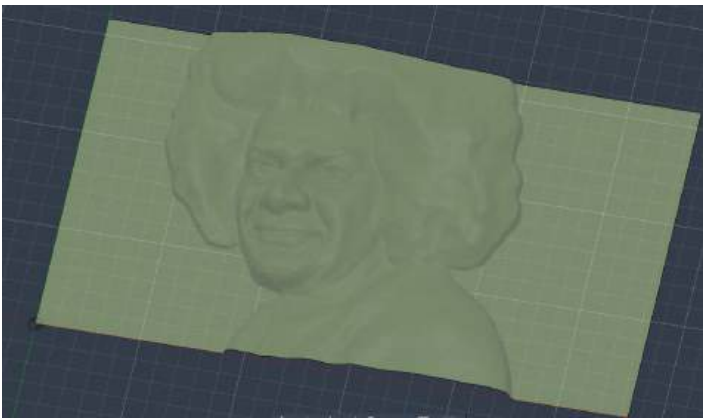
CNC Topographical Modeling



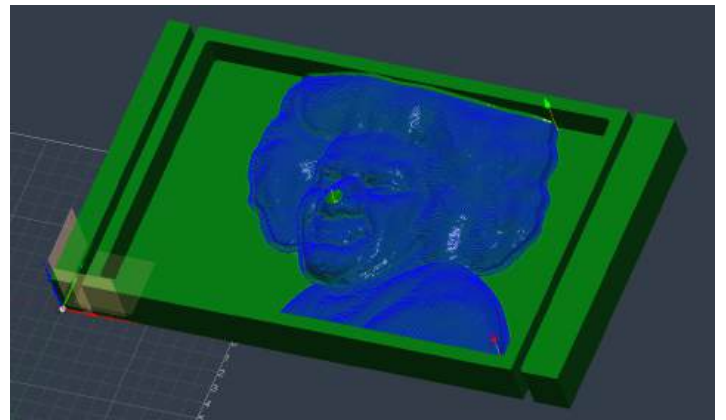
Source Image



Grayscale Gradient Heightmap



^ Converted Heightmap to Mesh ^



^ Created CNC Machining Toolpaths ^



^ Stock Preparation ^



^ CNC Router Operation ^



Final Product